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Energy deprivation to financial prosperity: Unveiling multidimensional energy Poverty's influence

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ABSTRACT

The purpose of this research is to assess how energy poverty affects an individual's social wellbeing and financial constraints. The present research examines the effects of energy deprivation on a family's well-being, educational attainment, and economic advancement using survey data from PSLM (Pakistan Social and Living Standard Measurement). The investigation's empirical findings imply that a household's needs for energy are directly correlated with the fuel type they use and how little or expensively they rely on renewable energy. The consequent implications of energy dependence on family prosperity are significant. Energy-poor households have been shown to be more prone to experience respiratory illnesses, pay more for health care, drop out of school, and have less employment opportunities.

1. Introduction

South Asian states have increased spending on education and health care but with a variation as a result of growth-induced more budgetary allocations. In order to ensure that the spending will be fruitful, some are also enhancing transparency in the provision of fundamental services. In Nepal, the state is devolving local communities more control over school administration. Non-governmental institutions have been hired to provide health services in Bangladesh and, in Pakistan. Although the majority of these initiatives have not yet undergone a thorough evaluation, early findings are promising. Therefore, the investigation of energy destitution issue in Pakistan is significant because the need for energy is expanding. According to NEPRA (2019) and ADB (2019), 17–18 % of the communities remain not linked to the national grid, and those that are statistically linked frequently endure blackouts with recurring and cycling power outages. It is impractical to ignore residential energy needs because access to energy is essential for maintaining subsistence conditions and raising living standards.

Many countries have achieved success in reducing the rate of hunger and extreme poverty. However, some countries indeed experience shocking levels of poverty. Some of these nations were unable to meet the ambitions set forth in the MDG (Millennium Development Goals). As a result, the SDGs (Sustainable Development Goals), which the United Nations associated with ending extreme poverty, continue to search for the most effective solutions to reduce poverty in terms of international as well as domestic interactions. Pakistan is a developing nation with inadequate capital to close the financial gap between investments and reserve reserves. Pakistan therefore heavily depends on foreign investment [1–4]. But in addition to the saving-investment imbalance, Pakistan's lack of human resources contributes to its economic instability.

Each SDG must be related to poverty and dropping the percentage of women, men, and children who are poor in half and adopting nationally adequate social support mechanisms and policies for all citizens by 2030, the first SDG seeks to "End poverty in all its forms everywhere."

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But the greatest impediment to getting sustainable development is poverty [5,6]. Analysis of multidimensional poverty highlights problems in the context of the SDGs, where eradicating poverty is among the first objectives to be achieved [7–11].

Pakistan has the second-highest population among South Asian states with a lower middle-income category. Since its origin, Pakistan's economy has encountered the vibrant patterns of neediness over the period and throughout space [12]. In a similar vein, factors like fast population increase and the evolution of the transportation and industrial sectors are over time increasing energy demand and worsening Pakistan's energy imbalance [13–15]. The achievement of SDG objective 7 in developing nations like Pakistan depends on having a clear understanding of two aspects of this target, particularly, "accessibility" and "affordability" to pure, reliable, advanced, and sustainable energy.

For the reason, this study explores the determinants of energy poverty in Pakistan by using PSLM survey data. In addition, this study investigates the impact of energy poverty on education, health and purchasing power of households using Amartya Sen's method of capability to examine the problem of energy poverty in Pakistan. This is the first ever research to incorporate the research gap in the previous literature. Since, the studies by Falak et al., 2014; Qurat-ul-Ann and Mirza, (2021); Li et al. [16]; Mahmood and Shah [17]; also measured energy poverty, however, these three categories are not explored in their researches for Pakistan. Hence, in order to analyze this measure within the context of the SDGs, this study focuses on four provinces of Pakistan. Furthermore, the rest of the exploration is organized as section 2 demonstrates the Literature review; section 3 explores methodology; section 4 displays results and discussion; and finally, section 5 ends up with the conclusion.

2. Literature review

South Asia accounted for the largest share of people worldwide who were living in poverty with multiple dimensions from 2010 through 2018. To be more precise, in 2017 almost half of the world's poorest people lived in South Asia. Economic expansion leads to greater inclusion [18–22]. In South Asia, there is a sorry history of misguided redistributive policies that have often been used as tools for electoral patronage. Samurdhi, the money exchange scheme in Sri Lanka, helped 44 % of the nation's impoverished residents but ignored 40 % of its impoverished citizens. Authorities are responding to the growing inequality-related push for reform. Samurdhi has undergone a revamp. India introduced a major program in 2006 that offers more than three months of work to any rural customer who is in distress. This well-known project, although far from perfect, has the advantage over others in that it is being watched over and evaluated, increasing the chances of its success.

Economic development is regarded as a high level concern, along with several both macro and micro challenges [10,11,23–27]. For the welfare of people, contemporary energy sources and access to clean are essential. According to Modi et al. (2006), the globe is experiencing extreme energy poverty. For cooking, about more than thirty percent of the world's populace relies on timber and other ineffective fuels including crop residues, dung, among others. Twenty five percent of the world's population does not have access to electricity, and the majority of them live in South Asia and Sub-Saharan Africa. These economies are more vulnerable due to the multifaceted effects of energy poverty [22].

Energy poverty in Indian communities has studied by Khandker et al. (2010). According to the survey, 20 % of individuals in metropolitan areas have low incomes and 28 % have low energy levels. The percentage of rural households who experienced energy poverty is 57 %, respectively. Unfortunate energy effectiveness in the warming/cooling of structures is one of the key causes of energy poverty. Betto et al. (2020) presented empirical support for the link between construction inefficiency and energy poverty employing the Italian Household Survey of 2018. Furthermore, Karpinska and Smiech (2020) provided evidence that inefficient buildings are the main factor contributing to energy poverty in nations throughout Europe.

Similarly, Tabata and Tsai (2020); Hu et al. [28]; Liu et al. [29]; Zhang et al. [30]; Mughal et al. [31]; and Ozughalu and Ogwumike (2019) offered empirical support for the theory that the level of education of the family head explains the household's energy poverty. New empirical data is introducing new causes of energy poverty to light. Predicting the possibility of energy poverty requires careful consideration of the attributes of the family head. According to earlier research, the level of energy poverty in a household is correlated with factors including literacy, employment position, and the sexual identity of the household head. Similarly, Tabata and Tsai [32]; Xin et al. [33]; Bhatti et al. [34]; Cao et al. [35]; Wang et al. [36]; and Ozughalu and Ogwumike (2019) offered empirical support for the theory that the level of education of the family head explains the household's energy poverty. New empirical data is introducing new causes of energy poverty to light. Financial inclusion has recently been identified as a key factor influencing energy poverty in Ghana Koomson and Danquah [37] and Turkey. For a considerable cohort in Australia, fuel poverty has also found to be connected to psychological well-being.

3. Methodology

The initial step in deciding the predominance of energy deprivation in Pakistan, more specifically, (Khyber Pakhtunkhwa (KPK), Baluchistan, Sindh, and Punjab) is to characterize what it is. This study has taken the data set (2003–2021) from PSLM. The likelihood indicator of multifaceted energy poverty for each affiliate of the family is constructed using the two different ideas of energy poverty that are adopted in this inspection: The main approach or idea is a multi-dimensional notion that reflects a number of patterns of consumption of energy and income levels within each cluster of quintiles. The main approach, however, makes the supposition that a household's provision of resources for its basket of goods is rational. Energy use is cited as being critical and one of the most significant elements of a family's income.

The upper class (5th quintile) is moreover remembered to have a little extent of energy spending to add up to burning through since power use is predicted to hit an immersion point as income increases, and the superior class likes to spend for the most part on extravagant commodities. The most unfortunate levels of income (either the first one or the second quintiles) have a more smaller extent of energy spending to whole use or pay since they don't want to burn through a lot on power and as opposed to depend on kindling or biomass for their regular necessities. The circumstances for the middle-income class (the third and fourth quintiles) are substantially different from those for the higher-income class (the 5th quintile) as well as the poor (the 1st and 2nd quintile). It is rationale to assume that if moderate-income categories of households are compelled to consume higher than 10 percent of their cash on energy, they will be starved of other essential services and commodities.

The equation used to express the aforementioned features of energy poverty is:

$$P(Y_i = 1) = J\,(Y_i = G_1) + J(\,Y_i = \{G_2\,, G_3, G_4\}\ \&\ S > 10\%) + J(Y_i = R_i) \tag{1}$$

$P(Y_i = 0)$, If non-energy destitute, where households in quintiles are G1-G4 spend the most on energy per person, and household uses biomass for food is represented by Ri. Whereas total household expenditure is represented by S. J(.) is a relation that measure on a valuation of 1 expecting the bracketed articulation is substantial, and 0 accepting regardless. Then, J(.) counterparts to 1 and the community of households would be seen as energy poor. In this way, energy poverty recurrence is:

$$EP_0 = \frac{1}{N}\sum_{i=1}^{n}[J(Y_i = G_1) + J(Y_i = \{G_2, G_3, G_4\} \,\&\, S > 10\%) + J(Y_i = R_i)] \quad (2)$$

Assuming once the energy poverty determinant is obtained, it is essential to investigate the energy poverty influences people's prosperity.

It's important to consider how energy poverty affects people's life after calculating the measure of energy poverty. This is a significant component of our study. Energy poverty (neediness of fuel) has been acquiring consideration from researchers and experts [38–40] since the fundamental investigation of Bradshaw and Hutton (1983). Energy destitution begins from low budget, absence of admittance to clean fuel or high energy costs.

4. Results and discussion

In addition, the consecutive cross-sections of the 'Multiple Indicator Cluster Survey' (MICS) statistics and PBS statistics are employed. The likelihood function is utilized to generate the factor 'energy poverty,' which depends on various models as illustrated in the estimation section and thought of energy destitution. On the off chance that a household is considered energy poor, the measurement 'energy poverty' expects the input 1; in any case, it utilizes the value 0 (See: Table 1). It is observed that per capita power use ought to be more than 1400 kWh person individual person annually or 117 kWh per individual per month in order for a person to cross the energy indigence threshold (Table 2).

The concentrate additionally examined the effect of energy neediness on the prosperity of families. The review relapsed wellbeing result, school result, and procuring result on the arrangement of exogenous factors as well as on energy neediness factors. It is accepted that energy destitution rely on the wellbeing factor. Thusly, energy desperation is instrumented on set of exogenous elements that are outstandingly associated with energy desperation, yet they are not related with error term, to ensure that energy destitution's constant can be adopted for inspection as it is unbiased. The revelations (See: Tables 3 and 4) suggested that energy desperation is vigorously linked with conventional fuel utilization of fuel wood, light oil for lighting, and in mix with lacking energy use as a result of 'extremeness'.

This concentrate firmly affirmed the effects of energy destitution on the well beings of families, including schooling (See: Table 3), health (See: Table 4) and earning capacity (See: Table 4) results. As far as the scale and extent of the effect of energy neediness on prosperity, energy destitution expands the anticipated likelihood of having a family part experiencing respiratory issue by practically 70pc compared to the non-energy-depleting category. This outcome focuses on wellbeing related respiratory sicknesses because of the utilization of customary biomass, like kindling for cooking as well as lamp fuel for lighting. The outcome further affirms that a child younger than 17 years under the energy-neediness bunch is probably going to be powerless to respiratory issues. The anticipated likelihood of being debilitated increments by 15 % in comparison with offspring of a similar age bunch in the non-energy-neediness bunch. A family's cash based spending is essentially identified with disorder, particularly respiratory issues. Compared to the population of the urban area, the numerous individuals typical of the rural region seem to increase the incidence of breathing problems.

Only families with university degrees have a favorable nexus with purchasing power (Table 4). This outcome is in line with the earlier Ethiopian study by Alem and Demeke (2020). Also, Legendre and Ricci (2015), households with higher levels of education make more energy-efficient decisions and stay out of energy poverty. Therefore, these findings vehemently advocated for having a completed university degree to influence gaining capacity. The evidence demonstrating that

Table 1
Variables description.

Factors	Details
Energy destitution	= 0 If there is no energy destitution and 1 otherwise
Family head	= 1 In the unlikely event that the homeowner is a male; else, 0.
School withdrawal	= 1 when a youngster is considered to have left schooling at the age of 7 through 17; alternatively, it represents 0.
Respiratory illness	= 1 in the event that a minimum of one family member is found to have the illness, and 0 in the other case.
Purchased a battery with cash	Family's battery usage over the previous 30 days
family-owned structures	= 0 in the event that the household does not own a single the personal computer, and 1 in the other case.
Energy/capita in kWh (annually)	Power utilization per person
Medical expenses	Medicinal usage throughout the family during the last four weeks.
Child	Younger than seventeen years of age
Log (per capita income)	The individual's income logarithms
Household size	Total person in family
The square of a child's typical work hours	Throughout a period within two weeks
LPG expenses	During thirty days
kerosene expenses	During thirty days
Television Possession	= 0 assuming that householder does not possess any TV; and 1 otherwise
Expenditure on Charcoal	Householder expenses on charcoal during thirty days
Square of power/person	kWh per year
Child Average employment	Immature biweekly work schedule
Rural	= 1 when rustic, and 0 in other case.
Householder possesses freezer/air conditioner	= 0 having not and 1 otherwise
Homemaker utilizing electric Fan	= 1 when having and 0 otherwise
The resident, having completed primary	= 1 in the event that they completed primary school, and 0 otherwise

Source: Liang et al. [41].

Table 2
Frequency of energy destitution in Pakistan (%).

Provinces	Energy Surplus	Energy Shortage
Sindh	24.5	75.5
Balochistan	42.1	57.9
Khyber Pakhtunkhwa	23.48	76.52
Punjab	15.29	84.71

Source: Authors own estimation

Table 3
Regression coefficient estimates for the medical outcome of "respiratory illness".

	DV (Respiratory illness)	OLS with Robust Standard Error [(a)]	2SLS with Robust Standard Error [(b)]
1	Energy poor	0.0398** (0.0178)	0.162*** (0.0521)
2	Medical expenses	0.0513*** (0.0150)	0.0390 (0.0306)
3	LPG expenses	0.0917** (0.0368)	0.152*** (0.0547)
4	Child	0.1084*** (0.0175)	0.0919*** (0.0439)
5	The square of a child's typical work hours	0.00320 (0.0189)	0.0521 (0.0342)
6	kerosene expenses	0.0486*** (0.0146)	0.0616** (0.0301)
7	Expenditure on Charcoal	0.135*** (0.0308)	0.292*** (0.0726)
8	The resident, having completed primary	−0.00554 (0.0299)	−0.0666 (0.0569)
9	Rural	−0.0987*** (0.0227)	−0.146* (0.0827)
10	Log (per capita income)	0.0372*** (0.00389)	0.0393*** (0.00858)
11	Constant	−0.3861*** (0.12809)	−0.9838*** (0.29431)
12	Obs.	4078	936
13	R^2	0.434	0.069

Note.
[a] F (10, 4055) = 15.86; Prob > F = 0.0000; Root MSE = 0.44971.
[b] 2SLS Wald $chi^2(9) = 64.17$; Prob > chi2 = 0.0000; Root MSE = 0.44043.

Table 4
Regression coefficient for 'Log (Income per Capita)'.

DV (Income per Capita)	OLS with Robust Standard Error [(a)]	2SLS with Robust Standard Error [(b)]
Household possesses Fan Consumption	0.0492***	0.056***
	0.0021	0.0031
Freezer Consumption	0.23***	0.2375***
	0.00320	0.00231
Air cooler	0.334**	0.0241**
	0.049	0.05201
Per capita power consumption (kWh)	0.0003***	0.0001***
	0.000	0.000
Householder (Incomplete Primary)	−0.073**	−0.220**
	0.0331	0.032
Householder (Literacy rate = 0)	−0.140***	−0.213***
	0.0029	0.00041
Householder Primary certification	0.040	0.0511
	0.751	0.022
Householder Upper Secondary certification	−0.220*	−0.179**
	0.0611	0.047
Household access to personal Computer	0.413**	0.612**
	0.051	0.0500
Householder Bachelor certification	0.406***	0.430***
	0.0001	0.002
energy poverty	−0.970***	−0.511***
	0.0023	0.00131
Household Strength	−0.923***	−0.0702***
	0.0067	0.0131
Rural	0.023	0.071
	0.39	0.49
Male householder	0.002	0.031
	0.018	0.101
Observations	3829	3829
R^2	0.41	0.39
Constant	0.2904*** (0.113)	0.2513*** (0.102)

Note.
[a] F (14; 3832) = 171.07; Prob. > F = 0.000; Root MSE = 0.413.
[b] 2SLS: Wald $Chi^2(9) = 26.93$; Prob.> $Chi^2 = 0.00$; Root MSE = 0.412.

energy destitution is unmistakably associated with energy availability and affordability is also highlighted by all regression results. This outcome is also in line with those of researchers.

5. Conclusion and policy implications

This study is the first of its nature in Pakistan to rigorously investigate the factors that lead to energy deprivation in the country. In this analysis, the concept of energy impoverishment in Pakistan is developed to illustrate the condition confronted by households with simultaneous access to and affordability of a different type of energy. The empirical results suggest that energy deprivation is a widespread problem throughout Pakistan. The impact of energy scarcity on household social wellbeing is also studied in the focus. The key findings suggested that, compared to a group of people who are not energy deficient, a household member who is in need of energy has a higher expected likelihood of contracting a respiratory illness. When compared to families who are not affected by energy neediness, the families who are affected by energy poverty will likely have a 48 percent lower earning capacity.

Children from communities who lack access to energy who are 17 years old and younger are more likely to be more vulnerable to and at a greater risk of becoming ill with respiratory illnesses than children from households who have access to energy. Additionally, contrasted with the non-energy-needy group, energy-helpless families may incur higher financial costs related to illness, particularly respiratory problems. More importantly, the research suggested that, assuming the typical Pakistani family consists of six individuals, the annual power usage should be above 583.3 kW h/household/month for a family to stay just above energy neediness line. The overall findings confirmed that the form of fuel consumption and access to energy are strongly connected to energy neediness.

Credit authors statement

All authors work equally and significant contribution in this draft.

Declaration of competing interest

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Data availability

Data will be made available on request.

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